CODEBOUND

An AnythingMCP Extension

Team CTRL-Z
Adharsh Siby Mathew Girija R Rohit Karunakaran Anish Saju

Solving the Large Dataset Bottleneck



The Challenge

44 MB

AI Inefficiency

Multi-step workflows fail when raw intermediate datasets (up to 44 MB) must travel back and forth through the AI model.

1 in 8

Production Cost & Latency

1 in 8 calls exceeds 50 KB, slowing down reasoning and inflating API costs.

CODEBOUND Approach

Code-to-Data Execution

Shift computation to where the data resides by turning versioned, typed JavaScript scripts into secure MCP tools.

Zero Raw Data Bloat

The sandbox processes large joins, filtering, and pagination locally; only the final compact result is returned to the AI model.

Target Metrics & Verification

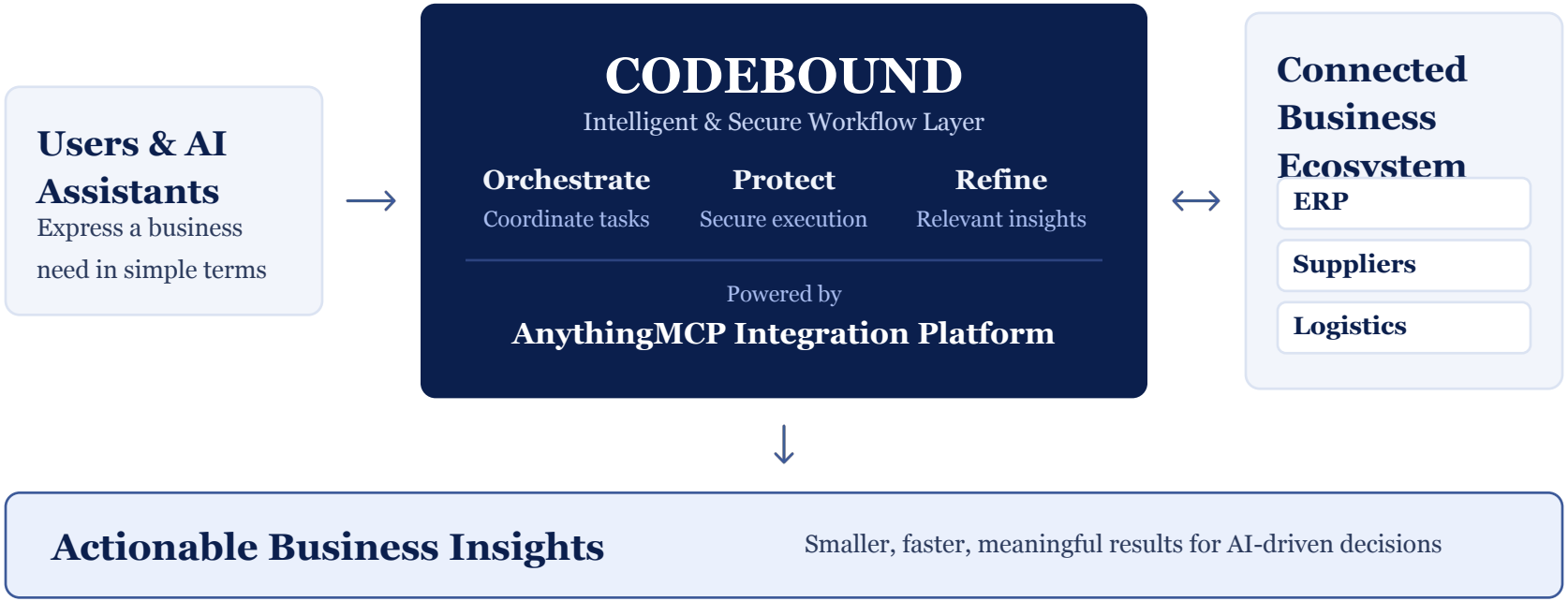
a. Lifecycle Management

Scripts can be dynamically created, edited, and executed entirely within the sandbox.

b. Isolated & Secure

Zero access to external networks, host mounts, or other customer data, with strict resource bounds (CPU, memory, and call limits).

Conceptual Architecture



Implementation

01 Docker Sandbox Runtime

- Scripts execute inside isolated, disposable Docker containers, separate from the core backend.
- Hard limits on execution time, memory and CPU prevent runaway loops and resource exhaustion.

02 Live-Authorizing Gateway

- Scripts cannot make arbitrary network calls.
- Requests to data sources or connectors pass through an audited, policy-enforcing gateway that checks workspace permissions and roles.

03 Typed JavaScript & Versioning

- Scripts are written with explicit types and version control, ensuring ensuring predictability and preventing untrusted code execution. execution.

04 Security & Attack Mitigation

- Validated against rigorous live attack frameworks.

Runtime: codebound-sandbox:mvp

CODEBOUND

Enterprise workflows, executed securely

CHALLENGES / MARKET FIT / SCALE / INTEGRATION / ROADMAP

01 Security & Authorization

- Docker sandboxes restrict network, CPU, memory and file access.
- Connector permissions are checked again during execution.

02 Market Fit

- Turn existing enterprise APIs into secure, actionable business workflows.

03 Scalability

- Separate business logic from execution; scale workers independently.

04 Easy Integration

- Add a secure workflow layer on top of existing systems—no replacement needed.

05 Future Roadmap

- Drag-and-drop visual workflow builder.
- AI-assisted creation of reviewable scripts.

THANK YOU